

magicmaus tutorial — maltose-binding protein (real data)

The [dummy walkthrough](#) showed the mechanics on 8 methyls. This one runs the same pipeline on a **real, full-size dataset**: the 370-residue *E. coli* maltose-binding protein (MBP), **192 methyls**, using experimental chemical shifts from **BMRB entry 7114** and the crystal structure **PDB 1ANF**. By the end you will have reproduced the headline result — magicmaus committing a single answer for all 192 peaks while never dropping the truth from its envelope — and the three-way comparison against MAGIC and MAUS on identical data.

Prerequisites: `.venv` with `python-sat` and `numpy` (see [README.md](#)). All commands run from the repo root.

1. The dataset

`examples/mbp/` ships the real inputs:

file	what it is
<code>1ANF.pdb</code>	MBP crystal structure
<code>hmqc.tsv</code>	192 methyl HMQC peaks (anonymous ids P1...P192) — the input
<code>hmqc_true.tsv</code>	answer key (BMRB assignment of each peak)
<code>noesy.tsv</code>	simulated 3D (H)CCH NOE network (nearest-8 within 8 Å), boolean
<code>hmqc_tentative.tsv</code>	same HMQC but with 24 peaks pre-anchored (used in §4)

192 methyls across 6 labelled types: Ala 44, Ile 22, Leu 60, Met 6, Thr 20, Val 40. That is a realistic large-protein methyl assignment problem.

2. First run — boolean NOESY

```
.venv/bin/python magicmaus.py \
  examples/mbp/1ANF.pdb examples/mbp/hmqc.tsv examples/mbp/
  noesy.tsv \
  --truth examples/mbp/hmqc_true.tsv --tol-h 0.01 --tol-c 0.05 \
  --out mbp_calls.tsv
```

```
methyIs(G nodes)=192  HMQC peaks=192  NOESY cross peaks=1650
NOE match (tol H+-0.01/C+-0.05): firm=502 ambiguous(dropped)=1148
unmatched=0
```

```
--- MAUS envelope (never excludes truth) ---
```

```
unique(1 option)      = 51/192
```

```
ambiguous(2-3 options)= 81/192
```

```
ambiguous(>3 options) = 60/192
```

```
--- magicmaus commitment (single coherent call) ---
```

```
confidence: unique=51 scored=67 ambiguous=74
```

```
--- scored vs truth ---
```

```
truth in MAUS option set = 192/192 = 100.0% (never-exclude
guarantee preserved)
```

```
magicmaus single call    = 121/192 = 63.0% correct
```

```
  unique    : 51/51 = 100.0% correct
```

```
  scored    : 44/67 = 65.7% correct
```

```
  ambiguous: 26/74 = 35.1% correct
```

Read it top to bottom:

- The SAT layer resolves **51/192 peaks uniquely** (all correct) and leaves 141 as multi-option sets — but **the truth is in the option set for all 192** (truth in MAUS option set = 100 %). That envelope guarantee never breaks.
- On top of that, magicmaus **commits a single answer for all 192 peaks: 63.0 % correct**, tiered by confidence (51 unique, 67 scored, 74 ambiguous). That is the number to beat with better data (§3–4).

The per-peak table

```
head -9 mbp_calls.tsv | column -t -s $'\t'
```

label	res_type	call	confidence	n_options	options
truth	call_correct	truth_in_set			
P1	L	L7CD2	scored	2	L7CD1,L7CD2
L7CD1	0	1			
P2	L	L7CD1	scored	2	L7CD1,L7CD2
L7CD2	0	1			
P3	V	V8CG1	ambiguous	2	V8CG1,V8CG2
V8CG1	1	1			
P4	V	V8CG2	scored	2	V8CG1,V8CG2
V8CG2	1	1			
P5	I	I9CD1	unique	1	I9CD1
I9CD1	1	1			

P6	I	I11CD1	unique	1	I11CD1
I11CD1	1	1			
P7	L	L20CD1	scored	2	L20CD1, L20CD2
L20CD1	1	1			
P8	L	L20CD2	ambiguous	2	L20CD1, L20CD2
L20CD2	1	1			

P5/P6 are pinned uniquely (unique, certain). P1/P2 are a Leu geminal pair the boolean network cannot orient, so magicmaus swaps them here — but note `truth_in_set = 1` for both: the right answer is still inside each envelope. This is exactly the boolean coin-flip the intensity column fixes next.

3. Add intensities

Real NOE cross peaks carry an intensity $\sim 1/r^6$ — the “how close” signal a boolean edge lacks. `make_intensity_noesy.py` reconstructs it for this dataset (from the truth key + structure), simulating a real experiment:

```
.venv/bin/python make_intensity_noesy.py examples/mbp/1ANF.pdb \
    examples/mbp/hmqc_true.tsv examples/mbp/noesy.tsv
    mbp_noesy_intensity.tsv

.venv/bin/python magicmaus.py examples/mbp/1ANF.pdb examples/mbp/
    hmqc.tsv \
    mbp_noesy_intensity.tsv --truth examples/mbp/hmqc_true.tsv \
    --tol-h 0.01 --tol-c 0.05 --soft-ambiguous
```

NOESY	single-answer correct	scored-tier correct
boolean	63.0 %	65.7 %
+ intensities	72.9 %	79.7 %
+ intensities --soft-ambiguous	79.7 %	85.8 %

`--soft-ambiguous` folds in the ambiguous NOE cross peaks MAUS discards as extra, intensity-weighted tie-breakers. Intensities lift the run **63 % → 80 %** — and the envelope stays at **100 %** the whole way. The scoring layer, idle on boolean data, does real work once the data carry information.

4. Two more levers

Tighter matching. Sharper shift tolerances make more NOEs resolve firmly, so more peaks land in the unique tier:

```
.venv/bin/python magicmaus.py examples/mbp/1ANF.pdb examples/mbp/
    hmqc.tsv \
    examples/mbp/noesy.tsv --truth examples/mbp/hmqc_true.tsv \
    --tol-h 0.01 --tol-c 0.02
# unique(1 option) = 70/192 ; single call = 156/192 = 81.2%
    correct
```

Tentative anchors. If you already know a few assignments, write them into the `res_type` cell of the HMQC file (e.g. L45D2 instead of L). `hmqc_tentative.tsv` carries 24 such anchors; they propagate through the NOE network and sharpen the rest:

```
.venv/bin/python magicmaus.py examples/mbp/1ANF.pdb \
    examples/mbp/hmqc_tentative.tsv examples/mbp/noesy.tsv \
    --truth examples/mbp/hmqc_true.tsv --tol-h 0.01 --tol-c 0.05
# unique(1 option) = 79/192 ; single call = 136/192 = 70.8%
    correct
```

24 anchors lift the unique tier from 51 to 79 — a few knowns go a long way.

5. The payoff — MAGIC vs MAUS vs magicmaus on identical data

`convert_to_magic.py` re-emits the intensity dataset as a MAGIC control bundle, so all three engines are scored on the **same** shifts, structure and NOE peaks. `score_three.py` runs the comparison:

```
.venv/bin/python convert_to_magic.py examples/mbp/hmqc.tsv \
    mbp_noesy_intensity.tsv examples/mbp/1ANF.pdb
    magic_run_intensity
# then run MAGIC on magic_run_intensity/control.txt (needs a ../
    magic checkout),
# and:
.venv/bin/python score_three.py
```

	methy1-level	residue-level	
truth-in-envelope			
MAGIC (scoring)	11/192 = 5.7%	20/192 = 10.4%	—
(no envelope)			
MAUS (SAT)	unique 51/192 = 26.6%		
192/192 = 100.0%			
magicmaus	140/192 = 72.9%	152/192 = 79.2%	
192/192 = 100.0%			
magicmaus +soft-amb	153/192 = 79.7%	164/192 = 85.4%	
192/192 = 100.0%			

Two facts the shared network makes plain:

- **MAUS cannot use intensity** — its constraints are boolean, so the 5th column changes nothing (identical 51 unique / 100 % envelope). It bounds the truth but stays undecided on the degeneracy.
- **MAGIC uses intensity but still lands at ~6–10 %** — global scoring over the full candidate space sits on a near-flat landscape and commits to a residue-level answer that cannot even resolve geminal pairs.

magicmaus runs the *same* intensity-weighted scoring MAGIC uses, but only inside MAUS's truth-containing domains — landing **13× MAGIC's methyl accuracy while keeping MAUS's 100 % envelope**. That is the synthesis: MAUS keeps the truth in reach, MAGIC's scoring extracts every bit of experimental signal to commit correctly within it. Neither half delivers this alone.

6. What to keep from a real run

On data without a truth key, drop `--truth` and read `mbp_calls.tsv`:

- **unique** calls — forced by hard constraints; trust them.
- **scored** calls — the NOE geometry's best pick; strong leads, verify where it matters.
- **ambiguous** calls — carry the *whole option set*, not the single call; these are genuine symmetries (mostly geminal pairs) awaiting stereo labelling, a 4D experiment, or more anchors.

The one invariant across every setting above: **the true methyl is never dropped from a peak's option set**. magicmaus commits when it can and abstains — visibly — when it cannot.

7. Beyond MBP — the five-protein benchmark

MBP is one of five targets. `make_peaklists.py` builds the same peak lists from any PDB + BMRB deposition, so the four de-novo blind targets of the MAUS paper (IL-2 and the Cas9 HNH/REC2/REC3 domains) drop straight in — see [../README.md#benchmark--five-real-bmr-b-targets](#) and each `examples/<target>/README.md`. Across all five the **100 % never-exclude envelope holds**, and magicmaus beats full-space MAGIC by up to ~10× while committing on every peak.